

Longitudes and Time Zones

Two important concepts, related to latitude and (especially) longitude are Local time (LT) and Universal time (UT)

Longitudes are measured from zero to 180° east and 180° west (or -180°), and both 180-degree longitudes share the same line, in the middle of the Pacific Ocean.

As the Earth rotates around its axis, at any moment one line of longitude—"the noon meridian"—faces the Sun, and at that moment, it will be noon everywhere on it. After 24 hours the Earth has undergone a full rotation with respect to the Sun, and the same meridian again faces noon. Thus each hour the Earth rotates by $360/24 = 15$ degrees.